

REGRESSION WITH A BINARY INDEPENDENT VARIABLE SUBJECT TO ERRORS OF OBSERVATION

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1. Introduction

In a recent study of the socio-economic effects of the disease bilharzia on the population of the Caribbean island of St. Lucia, Weisbrod et al. (1973) make abundant use of regression methods to analyze the various relationships of interest. A common independent variable in their work is a dummy variable that indicates the presence or absence of the disease in the person sampled (the observation unit). The nature of diagnosis for this particular disease is such that if a person is diagnosed as having the disease, he does indeed have it. However, if the diagnosis is negative there is a non-zero probability, η , that he has been diagnosed incorrectly and has the disease.

It is the intent of this note to consider the effects of such an independent variable – a binary variable subject to ‘errors of classification’ – in least squares regression. The results of both an analysis of bias in least squares parameter estimates and of the availability of alternative estimators is parallel to the classical case where both the variable and its measurement error are continuous random variables. Many details will not be repeated here in deference to the reader’s familiarity with that subject. The important practical difference between the two cases is that the information needed to obtain consistent (or nearly so) parameter estimates may be more readily available in the discrete case. In the study cited, for instance, extraneous information about η is available from patient medical histories and examination data.

The following section of the present article contains a brief exposition of the nature of a discrete ‘classification error’. Previous authors who have treated this material include Neyman (1950), Bross (1954) and Lord and Novick (1968). Sect. 3 then takes up an analysis of the effects of including an independent variable subject to such measure-

ment error in a multiple regression model, for which the paper by Cochran (1968) is a useful reference. This section also contains the presentation of a consistent technique for obtaining parameter estimates. A proof of consistency and details on precision measures for the method are contained in an Appendix.

2. Classification error in a binary variable

Let P be the true proportion of persons having the disease. If the i th element of the population is diagnosed as having the disease set $x_i = 1$; otherwise $x_i = 0$. The proportion of persons (in a population of N) being diagnosed as having the disease is $\tilde{P} = (1/N) \sum_{i=1}^N x_i$, and relates to P as follows. The proportion of persons diagnosed negative is $\tilde{Q} = (1 - \tilde{P})$. Of these, a proportion, η , have the disease. If, in addition, for persons diagnosed as having the disease (of which there are $N\tilde{P}$), some proportion, say ν , do not have it in fact, then in the total population, $N\tilde{P} - \nu N\tilde{P} + \eta(1 - \tilde{P})N$ actually have the disease. Therefore,

$$P = (1 - \nu)\tilde{P} + \eta\tilde{Q}. \quad (1)$$

In an 'errors-in-variables' framework, we let x , as defined above, be the observable random variable on a typical draw into the sample, and further define the variables X and u by the relation $x = X + u$. X is the 'true' state of the sampled person with respect to the disease, and u is a random measurement error. Obviously, the marginal distribution of X is Bernoulli with parameter P . Likewise, a Bernoulli distribution (with parameter $\tilde{P}$) characterizes the distribution of x . The joint distribution of (x, u) is obtained through simple manipulation, and is of particular interest in what follows. It appears in the form of a joint frequency table, below:

$\begin{array}{c} x \\ \backslash \\ u \end{array}$	0	1	$f(u)$
-1	$\eta\tilde{Q}$	0	$\eta\tilde{Q}$
0	$(1 - \eta)\tilde{Q}$	$(1 - \nu)\tilde{P}$	$(1 - \eta)\tilde{Q} + (1 - \nu)\tilde{P}$
1	0	$\nu\tilde{P}$	$\nu\tilde{P}$
$f(x)$	$\tilde{Q}$	$\tilde{P}$	1

(2)

Unlike the classical case, the measurement error in x does not have zero mean, nor is it uncorrelated with the true part. From (2), we see that ¹

$$E(u) = \nu\tilde{P} - \eta\tilde{Q} , \quad (3a)$$

$$V(u) = \nu\tilde{P} + \eta\tilde{Q} - (\nu\tilde{P} - \eta\tilde{Q})^2 , \quad (3b)$$

whereas,

$$C(x, u) = (\nu + \eta)\tilde{P}\tilde{Q} , \quad (3c)$$

a useful result for the next section.

3. Incorporation into the multiple regression model

Johnston (1963) considers the multiple regression model where every independent variable may be subject to random measurement error. We shall pursue one of his suggested techniques for obtaining consistent estimates of the regression coefficients, but we will suppose that of the several independent variables only x is subject to observation error. Extension to the more general case follows directly. Measurement error in x alone is, of course, sufficient to bias all least squares estimates of regression coefficients, not merely the coefficient on x . The exception is when x is orthogonal to all other independent variables.

For purposes of exposition we partition the population regression function as follows,

$$y = Z\gamma + X\beta + \epsilon , \quad (4)$$

where γ is a $[(k-1) \times 1]$ vector of unknown parameters, Z is an $[n \times (k-1)]$ matrix of observations on the 'other' independent variables, and ϵ is the $(n \times 1)$ vector of residuals with $E(\epsilon) = 0$ and $E(\epsilon\epsilon') = \sigma^2 I$. All variables are assumed to be measured as deviations from their respective mean values.

Least squares applied to the operational version of (4), namely

$$y = Z\gamma + x\beta + (\epsilon - u\beta) , \quad (5)$$

¹ Our notation is $V(\cdot)$ for the variance operator and $C(\cdot, \cdot)$ for covariance.

will result in the estimators

$$\begin{pmatrix} \hat{\gamma} \\ \hat{\beta} \end{pmatrix}_{LS} = (\tilde{X}'\tilde{X})^{-1} \tilde{X}'y, \quad (6)$$

where

$$\tilde{X}'\tilde{X} = \begin{bmatrix} Z'Z & Z'x \\ x'Z & x'x \end{bmatrix}, \quad \tilde{X}'y = \begin{bmatrix} Z'y \\ x'y \end{bmatrix}.$$

The accompanying sampling error in $(\gamma' \beta)'_{LS}$, e , will be of the form

$$e = \begin{pmatrix} \hat{\gamma} \\ \hat{\beta} \end{pmatrix}_{LS} - \begin{pmatrix} \gamma \\ \beta \end{pmatrix} = (\tilde{X}'\tilde{X})^{-1} \tilde{X}'\epsilon - (\tilde{X}'\tilde{X})^{-1} \tilde{X}'u\beta. \quad (7)$$

Assuming $\text{plim} \left(\frac{1}{n} \tilde{X}'\epsilon \right) = 0$, $\text{plim} \left(\frac{1}{n} \tilde{X}'\tilde{X} \right) = \Sigma_{\tilde{X}}$, and that u is asymptotically uncorrelated with the members of Z ,

$$\begin{aligned} \text{plim } e &= -\beta \text{plim} \left(\frac{1}{n} \tilde{X}'\tilde{X} \right)^{-1} \text{plim} \left(\frac{1}{n} \tilde{X}'u \right) \\ &= -\beta \begin{pmatrix} \bar{m}^{1x} & \times C(x, u) \\ \cdot & \cdot \\ \cdot & \cdot \\ \cdot & \cdot \\ \bar{m}^{k-1,x} & \times C(x, u) \\ \bar{m}^{xx} & \times C(x, u) \end{pmatrix} \end{aligned} \quad (8)$$

where $\bar{m}^{1x}, \dots, \bar{m}^{xx}$, are elements of the k th column of $\Sigma_{\tilde{X}}^{-1}$. (8) is just a special case of Cochran's equation (8.8) (see Cochran, 1968, p. 651).

If x is the only independent variable in the regression (with a constant term included), (8) reduces to

$$\text{plim } e = -\beta \left[\frac{C(x, u)}{V(x)} \right], \quad (9)$$

or, in terms of $\text{plim } \hat{\beta}_{LS}$:²

$$\text{plim } \hat{\beta}_{LS} = \beta \left[\frac{V(X) + C(X, u)}{V(x)} \right]. \quad (10)$$

Substituting relevant quantities, (10) is simply

$$\text{plim } \hat{\beta}_{LS} = \beta(1 - \nu - \eta), \quad (11)$$

or,

$$\text{plim } \hat{\beta}_{LS} = \beta(1 - \eta), \quad (12)$$

for the bilharzia example. To correct for least squares bias in general requires knowledge of both ν and η .

The direction of the proportionate bias is consistent with the usual result, that $\hat{\beta}_{LS}$ is biased downward. In the bilharzia case the proportionate bias is equal to the probability that a person diagnosed as *not* having the disease *does not* have it, i.e., the probability of correct diagnosis.³ Clearly, were η known, the estimator $\hat{\beta}_{LS}^* = \hat{\beta}_{LS}/(1 - \eta)$ would be consistent for β .

In the multiple regression model, though information about $C(x, u)$ may be available, and the remaining elements of (8) can – except for β – be estimated from $\frac{1}{n} \tilde{X}' \tilde{X}$, there is apparently no straightforward way of incorporating these quantities into the least squares procedure in order to correct for bias as there is in the two-variable case.

Given *knowledge* of $C(x, u)$, consistent estimators for γ and β can be obtained, however, either from maximum likelihood or from a procedure Johnston (1963, pp. 168–175) suggests. The method of instrumental variables may also be of practical value in this application.⁴ Below we concentrate on Johnston's technique, 'modified least squares', so named here for the resemblance of the equations determining these estimators to the normal equations of least squares.

For the model (4), letting $\xi = C(x, u)$, the modified least squares es-

² Again, cf. Cochran (1968) p. 651, eq. (9.1).

³ For the case in point, Weisbrod et al. (1973) feel that η may be as large as 0.5!

⁴ The instrument for x must be chosen to be correlated with the true part, X , and uncorrelated with both ϵ and u . There should be no difficulty finding an instrument for this problem, since the diagnosis is but one of many functions of a patient's symptoms.

timators are the $\hat{\gamma}$ and $\hat{\beta}$ given by:

$$\begin{pmatrix} \hat{\gamma} \\ \hat{\beta} \end{pmatrix} = \begin{bmatrix} M_{ZZ} & M_{Zx} \\ M'_{Zx} & (m_{xx} - \xi) \end{bmatrix}^{-1} M_{\tilde{X}y}, \quad (13)$$

where $M_{\tilde{X}y}$ is the k -element column vector of sample covariances $m_{1y}, \dots, m_{k-1,y}, m_{xy}$, with like interpretations for M_{ZZ}, M_{Zx} , and m_{xx} . Generally (13) is not equivalent to the equation set determining either the maximum likelihood estimators or the ordinary least squares estimators.⁵

The Appendix contains a consistency proof for the estimators and formulas for elements of their asymptotic covariance matrix. Due to their linearity they are asymptotically normally distributed. With regard to the matter of efficiency, if one is willing to adopt a mean-square-error (MSE) criterion for judging the virtues of competing estimators of β in the two-variable model, it is easy to show that circumstances may exist when the biased LS estimator will be preferred to the $\hat{\beta}$ of (13).

The asymptotic MSE of $\hat{\beta}$ is just its asymptotic variance, (A.10), which with the identities of sect. 2 inserted becomes

$$\text{MSE}(\hat{\beta}) = \frac{\sigma^2}{n\tilde{P}\tilde{Q}(1-\nu-\eta)^2}. \quad (14)$$

For the biased LS estimator, its asymptotic variance is $\sigma^2/n\tilde{P}\tilde{Q}$, so that

$$\text{MSE}(\hat{\beta}_{\text{LS}}) = \frac{\sigma^2}{n\tilde{P}\tilde{Q}} + \beta^2(\nu+\eta)^2, \quad (15)$$

⁵ However, for the two-variable case, where the equation determining β is

$$m_{xy} = (m_{xx} - \xi)\beta,$$

we find that

$$m_{xx}^{-1} m_{xy} = \beta(1 - \xi m_{xx}^{-1})$$

and

$$\text{plim}(m_{xx}^{-1} m_{xy}) = \beta \left[\frac{V(X) + C(X, u)}{V(x)} \right]$$

which is (10). That is, for the two-variable case, the modified least squares procedure coincides with ordinary least squares 'corrected' for bias. For more than two variables, however, the resemblance ends.

using (11). Then the (asymptotic) relative MSE efficiency of $\hat{\beta}$ is

$$\frac{\text{MSE}(\hat{\beta}_{\text{LS}})}{\text{MSE}(\hat{\beta})} = (1 - \nu - \eta)^2 [1 + t^2(\nu + \eta)^2] , \quad (16)$$

with $t^2 = \beta^2 n \tilde{P}\tilde{Q} / \sigma^2$. For certain combinations of values for $(\nu + \eta)$ and t^2 , the relative MSE efficiency of $\hat{\beta}$ can be less than one. Of course, for sufficiently large n (16) will exceed one whatever is the value of $(\nu + \eta)$.

It is difficult to assess the importance of these results in general since t^2 also depends on the model parameters, β and σ^2 . Nonetheless they serve to illustrate the point. At some 'large' sample size, the modified LS estimator for (13) will be superior to ordinary LS in MSE. Whether it compares favorably to maximum likelihood in asymptotic efficiency is unknown.

4. Conclusions

While the LS procedure in general gives biased and inconsistent parameter estimates in the errors-in-variables model, through a simple modification of the relevant moment matrix of independent variables consistent estimation of $(\gamma' \beta)'$ can be achieved. For our application, consistency depends upon a *knowledge* of ζ . Since the argument is on asymptotic grounds already, one might contemplate substituting a consistent estimate for ζ and proceeding. The difficulty here is the same one that plagues the classical 'extraneous estimation' literature. It is that as a practical matter the estimate for ζ will generally be obtained from a different population than that from which the n observations have been selected. With ζ estimated from such a source, therefore, it is not at all clear what sort of operation is implied by $\text{plim} [1 - m_{xx}^{-1}(\frac{1}{n}x'u)] [1 - \hat{\zeta}m_{xx}^{-1}]^{-1}$, which is central to the consistency proof. At this point the Bayesian would proceed directly to the 'product' sample space and claim consistency, while the strict classical statistician must be satisfied with a sort of 'approximate' consistency when $\hat{\zeta}$ is used in place of ζ .

While the maximum likelihood technique using the same a priori information is available, it may have little to recommend it over modified least squares save the possibility of large sample efficiency. Assuming normality for ϵ , the likelihood function for this model involves products of linear combinations of Normal densities, and therefore can be

expected to present some non-trivial computational problems.⁶ These same complexities in the sample likelihood make for a computationally difficult solution to an otherwise 'natural' Bayesian application.

Acknowledgement

This paper was completed with financial assistance from several sources. At the time of this writing the author held the position of Fulbright Research Scholar at the Center for Operations Research and Econometrics of the Université Catholique de Louvain, on leave from the University of Wisconsin. Support from the Graduate School of the University of Wisconsin and the National Science Foundation is gratefully acknowledged. J.B. Ramsey and M. Mouchart contributed helpful comments on the manuscript, though neither of them should be held responsible for any shortcomings that remain.

Appendix

1. Consistency of the estimators

The proof of consistency for the vector $(\hat{\gamma}' \hat{\beta})'$ of (13) is as follows. Write the equation set which determines $(\hat{\gamma}' \hat{\beta})'$ as

$$M_{Zy} = [M_{ZZ} M_{Zx}] \begin{pmatrix} \gamma \\ \beta \end{pmatrix} = M_{ZZ}\gamma + M_{Zx}\beta \quad (\text{A.1a})$$

and

$$m_{xy} = [M'_{Zx}(m_{xx} - \xi)] \begin{pmatrix} \gamma \\ \beta \end{pmatrix} = M'_{Zx}\gamma + (m_{xx} - \xi)\beta \quad (\text{A.1b})$$

⁶ There is, however, one feature of the model that is quite interesting when judged beside the maximum likelihood formulation of the classical errors-in-variables problem in regression. In such applications while the main parameters may be estimated consistently by maximum likelihood, the 'incidental' parameters (in this instance, the X_i 's) may not be (Kendall and Stuart, 1961, p. 384). This situation leads to biased variance estimators and the possibility that the conditions necessary for the maximum likelihood estimators to be BAN will not be met. Unlike the classical case, in the model under consideration here the number of nuisance parameters does not increase with sample size. In fact, the density function of each X_i is completely described by but three parameters, $\bar{P}$, η , and ν .

From (A.1b) we find

$$\beta = [m_{xx}^{-1}m_{xy} - m_{xx}^{-1}M'_{zx}\gamma] [1 - \xi m_{xx}^{-1}]^{-1}, \quad (\text{A.2})$$

which, when substituted into (A.1a) yields

$$\begin{aligned} M_{zy} = \frac{1}{n} Z' \left[I - x m_{xx}^{-1} \left(\frac{1}{n} \right) x' \{1 - \xi m_{xx}^{-1}\}^{-1} \right] Z\gamma + \\ + M_{zx} m_{xx}^{-1} m_{xy} [1 - \xi m_{xx}^{-1}]^{-1}, \end{aligned} \quad (\text{A.3})$$

where $\frac{1}{n} Z'Z = M_{ZZ}$, $\frac{1}{n} x'x = m_{xx}$, etc., as the equation set determining $\hat{\gamma}$. Setting $[I - x m_{xx}^{-1} (\frac{1}{n}) x' \{1 - \xi m_{xx}^{-1}\}^{-1}] = A$, and rearranging (A.3) gives

$$Z'Ay = Z'AZ\gamma,$$

or

$$\hat{\gamma} = (Z'AZ)^{-1} Z'Ay. \quad (\text{A.4})$$

For $\hat{\beta}$ we find (by substituting (A.4) into (A.2)):

$$\hat{\beta} = \frac{1}{n} m_{xx}^{-1} x' \{I - Z(Z'AZ)^{-1} Z'A\} y \{1 - \xi m_{xx}^{-1}\}^{-1}. \quad (\text{A.5})$$

Using $y = [Zx] [\gamma' \beta]' + [\epsilon - u\beta]$ in (A.4),

$$\hat{\gamma} = \gamma + (Z'AZ)^{-1} Z'A\epsilon + (Z'AZ)^{-1} Z'AX\beta, \quad (\text{A.6})$$

with $X = x - u$, as before. Taking probability limits,

$$\text{plim } \hat{\gamma} = \gamma + \text{plim} \left(\frac{1}{n} Z'AZ \right)^{-1} \text{plim} \left(\frac{1}{n} Z'AX\beta \right). \quad (\text{A.7})$$

But $AX = X - (1 - \xi m_{xx}^{-1})^{-1} x + (1 - \xi m_{xx}^{-1})^{-1} x m_{xx}^{-1} (\frac{1}{n} x'u)$. We note that $m_{xx}^{-1} (\frac{1}{n} x'u)$ is a scalar and that $\text{plim} (\frac{1}{n} x'u) = \xi$, and write $AX = X - x [1 - m_{xx}^{-1} (\frac{1}{n} x'u)] [1 - \xi m_{xx}^{-1}]^{-1}$. Now, $\text{plim} \frac{1}{n} Z'AX = \text{plim} \frac{1}{n} Z'X - \text{plim} \frac{1}{n} Z'x [1 - m_{xx}^{-1} (\frac{1}{n} x'u)] [1 - \xi m_{xx}^{-1}]^{-1}$, so that if $\text{plim} [1 - m_{xx}^{-1} (\frac{1}{n} x'u)] [1 - \xi m_{xx}^{-1}]^{-1} = 1$, $\text{plim} \frac{1}{n} Z'x = \text{plim} \frac{1}{n} Z'X$, thus $\text{plim} \frac{1}{n} Z'AX = 0^*$. Since

* The reader should be surprised if $E(u) = 0$ and u is specified as in sect. 2. But here variables are in 'deviations' form, hence $E(u) = 0$.

$\text{plim} \left(\frac{1}{n} x'u \right) = \zeta$, this will be true. Therefore, $\text{plim } \hat{\gamma} = \gamma$ and consistency is proved.

A similar development follows for $\text{plim } \hat{\beta}$. In this instance,

$$\begin{aligned} \text{plim } \hat{\beta} &= \beta \text{ plim } [1 - \zeta m_{xx}^{-1}]^{-1} \left[m_{xx}^{-1} \left(\frac{1}{n} x'X \right) \right] \\ &= \beta \text{ plim } [1 - \zeta m_{xx}^{-1}]^{-1} \left[1 - m_{xx}^{-1} \left(\frac{1}{n} x'u \right) \right] \\ &= \beta . \end{aligned} \tag{A.8}$$

2. Asymptotic covariance matrix

The large-sample variances and covariances for $\hat{\gamma}$ and $\hat{\beta}$ are easy to derive.

In the case of $\hat{\gamma}$, following (A.4),

$$\Sigma \hat{\gamma} = \frac{\sigma^2}{n} \text{plim} \left(\frac{1}{n} Z'AZ \right)^{-1} \left(\frac{1}{n} Z'AA'Z \right) \left(\frac{1}{n} Z'AZ \right)^{-1} . \tag{A.9}$$

Because A is not idempotent in this non-least squares situation, no further simplification of (A.9) is possible. The sample estimator for $\Sigma_{\hat{\gamma}}$ is obvious.

As for the variance of $\hat{\beta}$, from (A.5) we derive:

$$V(\hat{\beta}) = \frac{\sigma^2}{n} [V(x) - \zeta]^{-2} \text{plim} \left(\frac{1}{n} x'RR'x \right) , \tag{A.10}$$

where $R = I - Z(Z'AZ)^{-1}Z'A$ and is not idempotent. Again, $V(\hat{\beta})$ is easily estimated from sample information and a knowledge of the probabilities of misclassification.

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